# Sentiment Analysis Report

Amazon Consumer Product Reviews - NLP Capstone Project

## 1. Dataset Description

The sentiment analysis program uses the **Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products_May19.csv** dataset. The dataset contains 28,332 product reviews used by the program after selecting the **reviews.text** column. The review text provides natural-language customer opinions that can be analysed to determine whether the expressed sentiment is positive, negative, or neutral.

## 2. Text Preprocessing

Before sentiment analysis, the program removes rows with missing values in the **reviews.text** column using **dropna()**. Each review is then converted to a string, changed to lowercase, and stripped of leading and trailing whitespace. The text is processed with the **en_core_web_md** spaCy model. Stop words, punctuation, and whitespace tokens are removed using spaCy token properties such as **is_stop**, **is_punct**, and **is_space**. The remaining tokens are joined to produce cleaned text for sentiment analysis.

## 3. Sentiment Analysis Method

Sentiment is calculated using SpacyTextBlob in the spaCy pipeline. For each cleaned review, the program obtains a polarity score from **doc._.blob.polarity**. A polarity greater than zero is classified as Positive, a polarity below zero as Negative, and a polarity of zero as Neutral. Polarity scores closer to 1 indicate stronger positive sentiment, while scores closer to -1 indicate stronger negative sentiment.

## 4. Evaluation of Sample Results

| Review | Predicted sentiment | Polarity |
|---|---|---|
| 1 | Negative | -0.4500 |
| 2 | Negative | -0.5000 |
| 3 | Positive | 0.8000 |
| 4 | Positive | 0.5000 |
| 5 | Positive | 0.2500 |

The sample test produced three positive predictions and two negative predictions. Review 1, which describes an item as having bad quality and a missing component, was classified as Negative with a polarity of -0.4500. This result is consistent with the wording of the review. Reviews 3, 4, and 5 were classified as Positive, which is also consistent with phrases expressing happiness, better price, long-lasting batteries, and a great price.

Review 2 was classified as Negative with a polarity score of -0.5000. The review states that buying in bulk is a less expensive way to purchase the products. In context, this appears favourable because the customer is describing a cost advantage. This result therefore demonstrates that a polarity-based model may sometimes misinterpret words or phrases when their meaning depends on context.

## 5. Review Similarity

The program also compares the first two product reviews using spaCy's **similarity()** method. The resulting similarity score was **0.9519**. Because the score is close to 1, the model considers the two reviews highly semantically similar. This demonstrates how the **en_core_web_md** model can be used not only for text processing but also for comparing semantic relationships between review texts.

## 6. Strengths and Limitations

**Strengths:** The program provides a simple and repeatable method for processing large numbers of product reviews. It combines spaCy text processing with polarity-based sentiment analysis, produces clearly labelled sentiment categories and numerical polarity scores, and uses modular functions that make the code readable and reusable. The use of the medium spaCy model also enables semantic similarity comparisons.

**Limitations:** Polarity-based sentiment analysis does not always understand the full context of a customer statement. The classification of Review 2 illustrates this limitation. Removing stop words may also remove words that contribute to meaning in some sentences. In addition, the evaluation in this task is based on a small sample of reviews rather than a formal accuracy test against labelled sentiment classes. The similarity score should likewise be interpreted as the model's estimate of semantic similarity rather than proof that two customers expressed the same opinion.

## 7. Conclusion

The project successfully applies natural language processing to Amazon product reviews using spaCy and SpacyTextBlob. The program loads and cleans review text, predicts sentiment from polarity, tests the approach on sample reviews, and compares two reviews for semantic similarity. The sample results show that the approach can identify clear positive and negative language, while also demonstrating that context can cause misclassification. Overall, the project provides a practical example of both the usefulness and limitations of automated sentiment analysis.